

Systems Engineering Management Plan (SEMP)

Mission Systems Engineer

[Sept 29, 2025]

0.1 Introduction

0.1.1 Purpose

This SEMP defines the systems engineering approach for the Sentinel Bio+ mission, a constellation of six CubeSats designed to study space weather effects in low Earth orbit (LEO). This document establishes the technical management framework for all mission phases and provides integrated guidance for spacecraft design, integration, testing, operations, and decommissioning.

0.1.2 Scope

This SEMP covers all mission phases (Pre-Phase A through Phase F) and applies to:

- Spacecraft design and subsystem development
- Payload integration and interface management
- System integration and testing
- Launch operations and on-orbit commissioning
- Mission operations and data collection
- End-of-life disposal and mission closeout

0.1.3 Mission Overview

Mission Architecture:

- **Constellation:** Six 3U CubeSats
- **Deployment:** Planned for 2027
- **Orbit:** ~500 km altitude LEO
- **Inclinations:** 90°, 60°, and 40° (two satellites per inclination)
- **Mission Duration:** 2 years nominal operational lifetime

Science Objectives: Each satellite will measure:

- Radiation levels and energy spectra (100 keV to 25 MeV)
- Orbital drag and atmospheric density variations
- Magnetic field intensity and disturbances

Spacecraft Bus: TechEdSat-derived 3U CubeSat platform leveraging NASA Ames heritage, including:

- Standard 3U CubeSat structure
- Solar panels and Li-ion battery power system

- PC/104 avionics stack architecture
- Dual communications (UHF for T&C, S-band for science data)
- GPS receiver for timing and orbit determination

Primary Payload: Rila radiation dosimeter (80×70×30 mm, <100 g) providing 255-channel radiation spectra across 100 keV–25 MeV energy range, critical for understanding space radiation effects on satellites and crewed missions.

0.2 Applicable Documents

0.2.1 NASA Standards and Requirements

- NASA SP-2016-6105 Rev2: NASA Systems Engineering Handbook
- NPR 7123.1: Systems Engineering Processes and Requirements
- CubeSat Design Specification (CDS)

0.2.2 Mission-Specific Documents

- Sentinel Bio+ TechEdSat Integration Summary
- TechEdSat Heritage Documentation (TES-1 through TES-22)
- Requirements Compliance Matrix (RCM)
- Interface Control Document (ICD)
- Verification and Validation Matrix

0.3 Technical Planning

0.3.1 Systems Engineering Organization

SE Process Owner: Mission Systems Engineer
Responsibilities:

- Requirements definition and management
- Technical coordination across subsystems
- Interface management and control
- Risk identification and mitigation planning
- Technical review planning and execution
- Configuration management oversight

0.3.2 Engineering Toolset

- **Requirements Management:** Excel, JIRA
- **Design Tools:** SolidWorks (mechanical), KiCad (electrical)
- **Analysis:** MATLAB (orbit, power, thermal, link budget)
- **Configuration Management:** Git repositories for software/firmware
- **Data Management:** Centralized engineering repository

0.3.3 Key Milestones

- System Requirements Review (SRR)
- Preliminary Design Review (PDR)
- Critical Design Review (CDR)
- Test Readiness Review (TRR)
- Flight Readiness Review (FRR)
- Operational Readiness Review (ORR)

0.3.4 Enhancement for 2-Year Lifetime

Planning incorporates:

- TechEdSat heritage in long-duration avionics
- Enhanced thermal margins
- Redundant communications architecture
- Autonomous operations capabilities
- Battery cycle-life strategies for extended mission duration

0.4 Requirements Management

0.4.1 Requirements Flow-Down

Requirements are traced from stakeholder expectations through:

1. Mission-level requirements
2. System-level requirements
3. Subsystem specifications
4. Component requirements

0.4.2 Requirements Tracking

- **Tool:** Requirements Compliance Matrix (RCM)
- **Method:** Each requirement assigned unique identifier, verification method, and compliance status
- **Updates:** Requirements reviewed and updated at each technical milestone

0.4.3 Key Mission Requirements

Operational Life: Two-year operational lifetime supported by:

- Radiation-tolerant avionics
- Redundant communication paths
- Enhanced power management and battery balancing
- Watchdog/reset circuits

Data Collection: Continuous radiation monitoring with:

- 255-channel spectra resolution
- Energy range: 100 keV to 25 MeV
- Data collection every 15 minutes
- Per-satellite data volume: ~ 0.6 MB/day
- Constellation aggregate: ~ 3.6 MB/day

Reliability: Component selection and design practices ensuring:

- Radiation tolerance appropriate for 500 km LEO
- Battery cycling $< 10\%$ DoD per orbit nominal
- Thermal control maintaining Rila within -20°C to $+40^{\circ}\text{C}$

0.5 Interface Management

0.5.1 Interface Control Documentation

All interfaces documented in Interface Control Documents (ICDs) covering:

- Mechanical interfaces
- Electrical interfaces
- Data interfaces
- Thermal interfaces
- RF interfaces

0.5.2 Critical Interfaces

Payload-to-Avionics:

- Rila dosimeter: RS-422 serial interface (requires level-shifting transceiver to UART)
- ZES board: UART (3.3V), I²C, SPI interfaces
- Accelerometer and magnetometer: Digital interfaces to host PC
- Power: 12V (Rila), 5V (ZES and other payloads)

Communication Subsystems:

- UHF transceiver to antenna (monopole)
- S-band transceiver to antenna (patch or dipole)
- RF deck isolation from payload harnesses

Power Distribution:

- Solar panels to EPS
- EPS to battery pack
- Regulated 12V and 5V buses with load switches/current limiters
- Current limiter/load switch for 12V line (latch-up protection)

Internal Architecture:

- WiFi or TES-OS-style data handling for subsystem integration
- PC/104 stack interconnections
- GPS receiver to OBC for timing synchronization

0.5.3 EMI/EMC Management

- EMI isolation between Rila harness and RF decks
- Grounding plan to minimize interference
- Shielding provisions for sensitive electronics

0.6 Risk Management

0.6.1 Risk Identification and Tracking

Risks identified through design reviews, heritage mission lessons learned, and trade studies. Each risk assigned:

- Unique identifier
- Likelihood and consequence ratings
- Mitigation strategy
- Owner and closure criteria

0.6.2 Identified Risks and Mitigations

Risk	Impact	Mitigation
Thermal margin on Rila sensor	Measurement accuracy degradation	Add thermal straps and r
12V line latch-up events	Component damage, mission loss	Add external current limit
RS-422/UART compatibility	Payload data loss	Use level-shifting RS-422
Long-duration component degradation	Reduced mission life	Rad-tolerant avionics, wat
Battery cycle life	Power system failure	Battery balancing algorith
Communication link reliability	Data loss	Dual-path comms (S-band
Single-point failures	Mission loss	Redundancy in critical fun

0.6.3 Risk Review Cadence

- Risks reviewed at each technical milestone
- Monthly risk review meetings during integration and test
- Post-launch risk monitoring during commissioning

0.7 Configuration Management

0.7.1 Configuration Control

All hardware, software, and documentation under formal configuration control:

- **Hardware:** Part numbers, revision levels, serial numbers tracked
- **Software/Firmware:** Git version control with tagged releases
- **Documentation:** Revision-controlled with change logs

0.7.2 Change Control Board (CCB)

- **Composition:** Mission Systems Engineer (chair), subsystem leads, project management
- **Authority:** Approves all changes to baselined configurations
- **Process:** Change requests evaluated for technical impact, cost, and schedule

0.7.3 Baseline Configuration Items

- TechEdSat heritage software (TES-OS, DTN, Cognitive Engine-1) as candidate elements
- Rila dosimeter firmware (factory-calibrated)
- ZES board firmware with EDAC and TMR FRAM

- OBC flight software
- Ground system software

0.7.4 Configuration Audits

- Functional Configuration Audit (FCA) post-environmental testing
- Physical Configuration Audit (PCA) pre-delivery
- As-built documentation maintained throughout mission life

0.8 Technical Data Management

0.8.1 Data Repository Structure

Centralized engineering repository containing:

- CAD files (SolidWorks assemblies and parts)
- Electrical schematics (KiCad projects)
- Analysis results (MATLAB scripts and outputs)
- Test reports and procedures
- Requirements and interface documents
- Review presentation materials

0.8.2 Data Backup and Archival

- **Operations Data:** Backed up daily, stored in cloud
- **Engineering Data:** Version-controlled, redundant storage
- **Science Data:** Stored with metadata, archived for long-term access

0.8.3 Cloud-Based Ground Integration

Integration with cloud-based ground station services:

- AWS Ground Station as a Service (GSaaS) for distributed operations
- Iridium SBD backup for emergency T&C
- Cloud demodulation and data processing (TES-13 framework heritage)

0.9 Technical Assessment

0.9.1 Technical Review Process

Reviews conducted per NASA lifecycle at key milestones to assess technical maturity, identify risks, and authorize progression to next phase.

0.9.2 Review Definitions and Success Criteria

System Requirements Review (SRR):

- **Purpose:** Confirms system requirements are complete, consistent, technically feasible, and meet stakeholder needs
- **Exit Criteria:** Requirements baselined, mission concept validated, preliminary architecture defined

Preliminary Design Review (PDR):

- **Purpose:** Evaluates preliminary design for each system element to ensure requirements satisfaction with acceptable risk
- **Exit Criteria:** Preliminary design mature, interfaces defined, verification approach established

Critical Design Review (CDR):

- **Purpose:** Confirms detailed design is complete and ready for fabrication, integration, and testing
- **Exit Criteria:** Final design drawings released, all ICDs complete, procurement initiated

Test Readiness Review (TRR):

- **Purpose:** Assesses readiness to proceed into formal testing of hardware/software
- **Exit Criteria:** Test procedures approved, test hardware ready, facilities verified

Flight Readiness Review (FRR):

- **Purpose:** Verifies space vehicle and support systems are ready for launch and flight
- **Exit Criteria:** All testing complete, anomalies resolved, launch operations plan approved

Operational Readiness Review (ORR):

- **Purpose:** Ensures mission operations team and ground systems are prepared for on-orbit operations
- **Exit Criteria:** Ground systems operational, flight procedures validated, operations staff trained

0.9.3 Additional Assessment Focus

- Verification of autonomy (DTN, UIS scheduling)
- Resilience validation for two-year lifespan
- Long-duration operational scenarios
- Radiation effects on avionics and payload performance

0.10 Decision Analysis

0.10.1 Trade Study Process

Formal trade studies conducted for significant design decisions using:

- Clearly defined evaluation criteria
- Weighted scoring methodology
- Quantitative and qualitative assessments
- Documentation of assumptions and rationale

0.10.2 Completed Trade Studies

Radiation Sensor Placement:

- **Criteria:** Thermal environment, EMI exposure, volume allocation, field of view
- **Decision:** Mounted in upper 0.2U section with thermal straps
- **Rationale:** Minimizes thermal variation, reduces EMI from RF decks

Communication Antenna Configuration:

- **Criteria:** Gain pattern, deployment reliability, volume, mass
- **Decision:** UHF monopole deployable, S-band patch/dipole fixed
- **Rationale:** Balances link performance with mechanical simplicity

Downlink Scheduling Strategies:

- **Criteria:** Power usage, data latency, ground station availability, autonomy
- **Decision:** S-band 6 min/orbit typical, scheduled during sun passes, cognitive scheduling
- **Rationale:** Optimizes power budget while meeting data volume requirements

Exo-Brake Integration:

- **Criteria:** Mass budget, deorbit reliability, disposal compliance, TRL
- **Decision:** Incorporate Exo-Brake or drag device for end-of-mission disposal
- **Rationale:** Ensures controlled reentry compliance, leverages TES-3/10 heritage

0.10.3 Key Trade Criteria

- Power consumption
- Volume and mass
- Technology Readiness Level (TRL)
- Cost and schedule impact
- Heritage and reliability

0.11 Verification and Validation (V&V)

0.11.1 Verification Approach

Methods:

- **Test:** Physical testing to demonstrate compliance
- **Analysis:** Engineering calculations and simulations
- **Inspection:** Visual or dimensional verification
- **Demonstration:** Functional operation under nominal conditions

0.11.2 Verification Matrix

All requirements mapped to verification methods in V&V matrix, including:

- Requirement identifier and statement
- Verification method(s)
- Test procedure reference
- Verification status and completion date

0.11.3 Validation Activities

Mission Operations Simulation:

- End-to-end mission scenario testing
- Ground system compatibility validation
- Autonomous operations verification
- Fault recovery demonstrations

Radiation Spectrum Validation:

- Ground calibration using Rilas radiationless method (light pulsebased)

- Comparison with accelerator testing baseline ($\pm 15\%$ accuracy)
- Cross-validation between constellation satellites post-launch

Long-Duration Testing:

- Accelerated life testing for avionics and batteries
- Extended thermal cycling
- Communication link stress testing
- Battery cycle-life validation (shallow DoD profiles)

0.11.4 Environmental Testing

- Vibration (qualification and acceptance levels)
- Thermal vacuum (operational and survival temperatures)
- EMI/EMC testing
- Functional testing pre- and post-environmental exposure

0.12 Product Integration and Transition

0.12.1 Integration Sequence

Subsystems integrated sequentially within 3U frame:

Phase 1: Avionics Stack (0.8U)

- Onboard computer installation
- EPS integration
- PC/104 stack assembly and functional test

Phase 2: Communications (0.7U)

- S-band transceiver integration
- RF deck installation
- Antenna mounting and deployment test

Phase 3: Payload Integration (1.5U total)

- Rila dosimeter mounting (0.2U) with thermal interface
- ZES board installation with RS-422/UART transceiver
- Radiation-tolerant host PC (1.01.3U)
- Accelerometer and magnetometer integration

Phase 4: Power and Thermal

- Battery pack installation
- Solar panel attachment
- Thermal hardware (straps, radiators)
- Harness routing and strain relief

Phase 5: Final Assembly

- GPS receiver integration
- Final harness connections
- EMI shielding installation
- Closeout and mass properties measurement

0.12.2 Integration Testing

Post-integration functional and interface testing:

- Power system checkout (all voltage rails)
- Communication link verification (UHF and S-band)
- Payload data collection test
- GPS lock and timing validation
- End-to-end command and telemetry test

0.12.3 Transition Criteria

Authorization to proceed to launch operations requires:

- Successful completion of all environmental testing
- Flight software validation
- Successful downlink test with ground stations
- FRR completion with all action items closed
- Launch site safety review approval

0.12.4 Autonomous Integration Features

- TES-OS middleware for simplified subsystem communication
- Cloud-demodulation hooks for operations handover
- Autonomous fault detection and recovery capabilities

0.13 Operations and Sustainment

0.13.1 Mission Operations Architecture

Ground Segment:

- **Primary:** Atlas Space Ops ground stations (S-band science downlink)
- **Secondary:** UHF ground stations (T&C)
- **Backup:** Iridium SBD link for emergency command

Operations Concept:

- Daily operations planning
- Autonomous scheduling with cognitive communications
- Store-and-forward data handling via Delay Tolerant Networking (DTN)
- Cloud-based data processing and distribution

0.13.2 Communication Links

S-band (Science Data):

- High data rate: supports ~ 3.6 MB/day constellation aggregate
- Downlink windows: 6 min/orbit typical, scheduled during sun passes
- Modulation optimized for power efficiency

UHF (Telemetry & Command):

- Low data rate, continuous or scheduled
- Command uplink and spacecraft health monitoring
- Emergency safing capability

Iridium SBD (Backup T&C):

- Emergency command capability
- Status beacons during contingency scenarios
- Supports global coverage independent of ground station network

0.13.3 Power Management Strategy

Battery Management:

- Target: <10% Depth of Discharge (DoD) per orbit nominal
- Shallow cycling for long life: 510% DoD typical
- Cell balancing during sun passes
- Temperature maintained 030rC
- Maximum State of Charge (SOC) limited to 9598%

Operational Power Budget:

- Base load (no S-band TX): ~2.75 W
 - OBC + avionics: 1.5 W
 - EPS overhead: 0.3 W
 - UHF T&C: 0.2 W average
 - GPS: 0.2 W average
 - ZES + Rila: 0.55 W continuous
- S-band downlink: +8.0 W during 6-min TX window

Energy Balance:

- BOL: Net positive ~0.47 Wh/orbit with one downlink in sun
- EOL: Net neutral or slightly negative; mitigated by reduced TX cadence or improved sun-pointing
- Daily margin (BOL): ~38 Wh/day available for contingencies

Eclipse Operations:

- Battery discharge: ~1.38 Wh nominal (5.7% DoD)
- Worst case (S-band in eclipse): 2.18 Wh (9.1% DoD)
- Strategy: Schedule high-rate downlinks in sun passes where possible

0.13.4 Data Collection and Handling

Radiation Data:

- OBC polls Rila every 15 minutes
- Dataset size: $\sim 6,242$ bytes per collection
- Daily volume per satellite: ~ 0.6 MB
- 255-channel spectra across 100 keV–25 MeV

Science Data Processing:

- On-board: Initial quality checks, data compression
- Ground: Full spectrum analysis, cross-satellite correlation
- Archival: Long-term storage with metadata for scientific community access

0.13.5 Autonomous Operations

- Cognitive communications for optimized ground station scheduling
- Autonomous fault detection and safing
- Watchdog timers and reset circuits for recovery from SEU events
- Store-and-forward using DTN protocols

0.13.6 Mission Duration and Sustainment

- **Nominal Mission:** 2 years
- **Extension Scenario:** Up to 5 years possible with degraded performance
- **Sustainment Activities:**
 - Battery health monitoring and management
 - Software patch capability
 - Trending analysis for predictive maintenance
 - Periodic calibration checks

0.14 Closeout

0.14.1 End-of-Mission Operations

Decommissioning Sequence:

1. Final science data collection campaign
2. Complete data downlink and verification

3. Battery safing procedures
4. Exo-Brake or drag device deployment for controlled reentry
5. Final telemetry collection
6. Transmitter shutdown

Disposal Compliance:

- LEO CubeSat disposal practices per debris mitigation guidelines
- Exo-Brake deployment ensures reentry within 25 years (heritage from TES-3, TES-10)
- Passive deorbit via atmospheric drag if Exo-Brake not deployed

0.14.2 Data Archival

- Complete telemetry archive
- Full science dataset with calibration data
- Engineering performance data
- Lessons learned documentation

0.14.3 Final Analysis and Reporting

- Mission performance assessment
- Science data quality evaluation
- Subsystem performance analysis
- Anomaly investigation reports
- Technology demonstration results
- Final mission report

0.14.4 Knowledge Capture

- Design and operational lessons learned
- Heritage documentation for future missions
- Technology maturation assessment
- Contribution to TechEdSat program knowledge base

.1 TechEdSat Heritage Adaptations for Sentinel Bio+

The Sentinel Bio+ mission leverages extensive TechEdSat heritage to achieve a reliable two-year constellation mission. Key adaptations include:

.1.1 Communications Architecture

Heritage: TES-3, TES-13

- Dual-path approach: S-band for high-rate science data, Iridium SBD backup for emergency T&C
- Cloud-based demodulation using AWS GSaaS framework (TES-13)
- Reduces operator workload and enables global coverage
- Proven reliability in LEO environment

.1.2 Autonomous Operations

Heritage: TES-11 Cognitive Engine-1

- TES-OS concepts for middleware and subsystem integration
- Cognitive communications for optimized ground station scheduling
- Autonomous fault detection and recovery
- Reduces ground station contact requirements and operational costs

.1.3 Ground System Integration

Heritage: TES-13 AWS framework

- Cloud-based data processing and distribution
- Scalable architecture supporting six-satellite constellation
- Low-cost global coverage through commercial ground networks
- Automated scheduling and mission planning tools

.1.4 Deorbit Strategy

Heritage: TES-3, TES-10 Exo-Brake

- Incorporation of Exo-Brake or drag augmentation device
- Ensures controlled reentry and debris mitigation compliance
- Proven technology with multiple successful demonstrations
- Enables responsible end-of-mission disposal

.1.5 Radiation Tolerance and Reliability

Heritage: TES-7, TES-10 omni-board

- Rad-tolerant avionics with watchdog circuits
- SEU mitigation through EDAC (Error Detection and Correction)
- TMR FRAM for critical data storage
- ZES100 current limiter for latch-up protection
- Demonstrated reliability in LEO radiation environment

.1.6 Power and Thermal Management

Heritage: TES-10 extended operations

- Battery cycle-life strategies for long-duration missions
- Thermal control using radiators and thermal straps
- Power management algorithms optimized for shallow DoD cycling
- Proven thermal design for payload temperature stability

.1.7 System Integration Approach

Heritage: TES program PC/104 architecture

- Modular avionics stack for flexibility and maintainability
- Standard interfaces reduce integration complexity
- Proven mechanical and electrical integration processes
- Enables rapid subsystem replacement if needed

This heritage ensures Sentinel Bio+ leverages proven NASA smallsat practices while meeting its unique requirement for a two-year constellation mission studying space weather effects in LEO.

.2 Payload and Subsystem Specifications

.2.1 Rila Radiation Dosimeter

Developer: Space Research and Technology Institute, Bulgarian Academy of Sciences
Physical Specifications:

- Dimensions: $80 \times 70 \times 30$ mm (~ 0.17 U volume)
- Mass: <100 g

- Mounting: Aluminum bracket with thermal interface

Electrical Specifications:

- Input Voltage: 12V DC
- Power Consumption: 0.35 W nominal, <0.5 W peak
- Interface: RS-422 serial
- Data Rate: 56.8 or 115.2 kb/s
- Data Format: Structured frames, ~6,242 bytes per 15-min collection

Performance Specifications:

- Energy Range: 100 keV to 25 MeV
- Resolution: 255-channel dosimetry
- Dynamic Range: 250 (extendable to 1000)
- Calibration Accuracy: $\pm 15\%$ compared to accelerator testing
- Calibration Method: Radiationless (light pulsebased)

Environmental:

- Operating Temperature: -20°C to $+40^{\circ}\text{C}$
- Survival Temperature: -40°C to $+60^{\circ}\text{C}$
- Radiation Qualification: LEO environment at 500 km
- Heritage: Liulin series on ISS and Ten-Koh satellite
- Expected Lifetime: ≥ 2 years at 500 km LEO, up to 5 years possible

.2.2 ZES Board (ZSOM-M01 SOM + NS01A)

Function: Middleware between payloads and OBC

Specifications:

- Input Voltage: 5V
- Interfaces: UART (3.3V), I²C, SPI
- Radiation Mitigation: EDAC, TMR FRAM
- Protection: ZES100 current limiter for SEL protection
- Processing: Handles data formatting and protocol conversion

.2.3 Supporting Sensors

Accelerometer:

- Type: Board-level MEMS sensor
- Purpose: Measures orbital drag and spacecraft dynamics
- Interface: I²C or SPI digital
- Power: <0.1 W

Magnetometer:

- Type: Board-level 3-axis sensor
- Purpose: Measures magnetic field intensity and variations
- Interface: I²C or SPI digital
- Power: <0.1 W

GPS Receiver:

- Purpose: Position, velocity, and timing
- Interface: Serial (UART)
- Power: 0.2 W average (duty-cycled)

.2.4 Volume Allocation (3U Frame)

Subsystem	Volume (U)	Notes
Avionics, EPS, OBC	0.8	PC/104 stack
S-band transceiver + RF deck	0.7	Includes RF components
Rila dosimeter	0.2	Upper section placement
Host PC (rad-tolerant)	1.01.3	Processing and payload interface
Accelerometer, magnetometer	Included in host PC section	Board-level integration
Subtotal	2.73.0	
Harness, fasteners, thermal	0.00.3	Margin for integration
Total	3.0	Exo-Brake omitted

.3 Parts and Components List

.3.1 Structure and Bus

- 3U CubeSat frame and rails (aluminum)
- PC/104 avionics stack system
- Mounting brackets and standoffs for payloads
- Fasteners (stainless steel, titanium)

.3.2 Avionics

- Onboard computer (TES-OS compatible or equivalent)
- Power distribution board (EPS with MPPT)
- GPS receiver (commercial space-grade)
- RS-422/UART level shifter board

.3.3 Power System

- Deployable or body-mounted solar panels (triple-junction)
- Li-ion battery pack (~ 30 Wh, 3U-class)
- Load switches/current limiters for 12V and 5V rails
- Battery management system with balancing

.3.4 Communications

- UHF radio transceiver (T&C)
- S-band transceiver (science data downlink)
- UHF monopole deployable antenna
- S-band patch or dipole antenna (fixed)
- Iridium SBD modem (backup T&C, optional)

.3.5 Payloads

- Rila dosimeter ($80 \times 70 \times 30$ mm, < 100 g)
- Radiation-tolerant host PC
- Accelerometer (board-level MEMS)
- Magnetometer (board-level 3-axis)
- ZES ZSOM-M01 SOM + NS01A board

.3.6 Cabling and Support

- Wire harness (UART, RS-422, power distribution)
- Thermal hardware (heat straps, radiator panels)
- EMI shielding and grounding components
- Conformal coating materials (moisture protection)

.4 Acronyms and Definitions

CCB Change Control Board

CDR Critical Design Review

CDS CubeSat Design Specification

DoD Depth of Discharge

DTN Delay Tolerant Networking

EDAC Error Detection and Correction

EMC Electromagnetic Compatibility

EMI Electromagnetic Interference

EPS Electrical Power System

FCA Functional Configuration Audit

FRR Flight Readiness Review

GSaaS Ground Station as a Service

ICD Interface Control Document

LEO Low Earth Orbit

MPPT Maximum Power Point Tracking

OBC Onboard Computer

ORR Operational Readiness Review

PCA Physical Configuration Audit

PDR Preliminary Design Review

RCM Requirements Compliance Matrix

RF Radio Frequency

SBD Short Burst Data

SEMP Systems Engineering Management Plan

SEL Single Event Latchup

SEU Single Event Upset

SOC State of Charge

SRR System Requirements Review

T&C Telemetry and Command

TES TechEdSat

TMR Triple Modular Redundancy

TRL Technology Readiness Level

TRR Test Readiness Review

UHF Ultra High Frequency

UART Universal Asynchronous Receiver-Transmitter

V&V Verification and Validation